complaint-router

Routing consumer financial complaints to the right team, with a system that keeps learning

KSE MLOps · Lab 1 · Zakhar Staryk · github.com/goodfrog33/kse-mlops-complaint-router

18.5 M
complaints in the CFPB database

3.8 M
with complaint text

10
product queues

At a glance

Problem

- ~21,000 complaints a day to the US regulator CFPB
- Each must reach the team that owns the product

System

- Predicts 1 of 10 product queues from the complaint text
- Confident → auto-routed; otherwise → human
- Retrained on drift, monitored before labels arrive

Data

- CFPB Consumer Complaint Database (public)
- 3.8 M complaint texts, 2015–2025
- Replayed month by month

Fit with the 5 labs

- Yes: pipeline, LLM, serving and agent all have real material
- Caveat: no new complaint text after 2025

Every complaint has to reach the team that owns it



~21,000 / day

complaints in 2026 (measured)

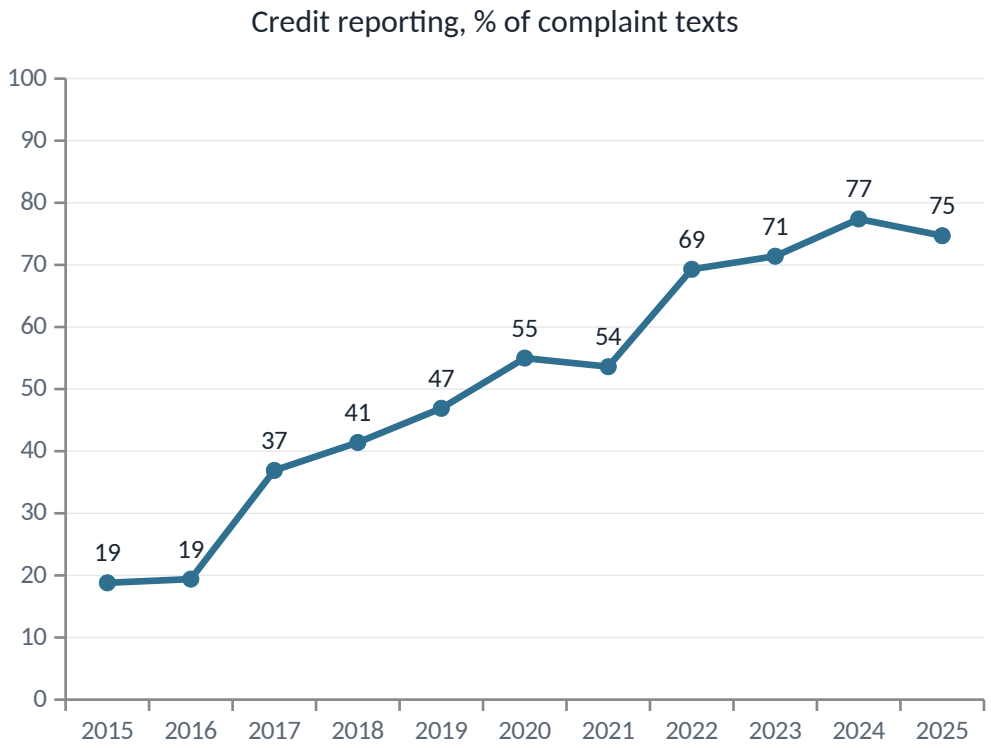
15 days

response window per complaint

Systematic

errors hide until labels return

The data keeps moving, so a one-off notebook would decay



2 taxonomy changes

2017-04 and 2023-08

19% → 77%

credit reporting share

30× volume

0.17 M → 5.4 M a year

Text stopped

CFPB, 2026

Success is measured on three separate layers

Business

≤ 8%
routed to the wrong queue

≥ 60%
routed without a human

Model

≥ 0.80
macro-F1

≥ 0.95
precision on auto-routed

System

≤ 300 ms
p95 latency

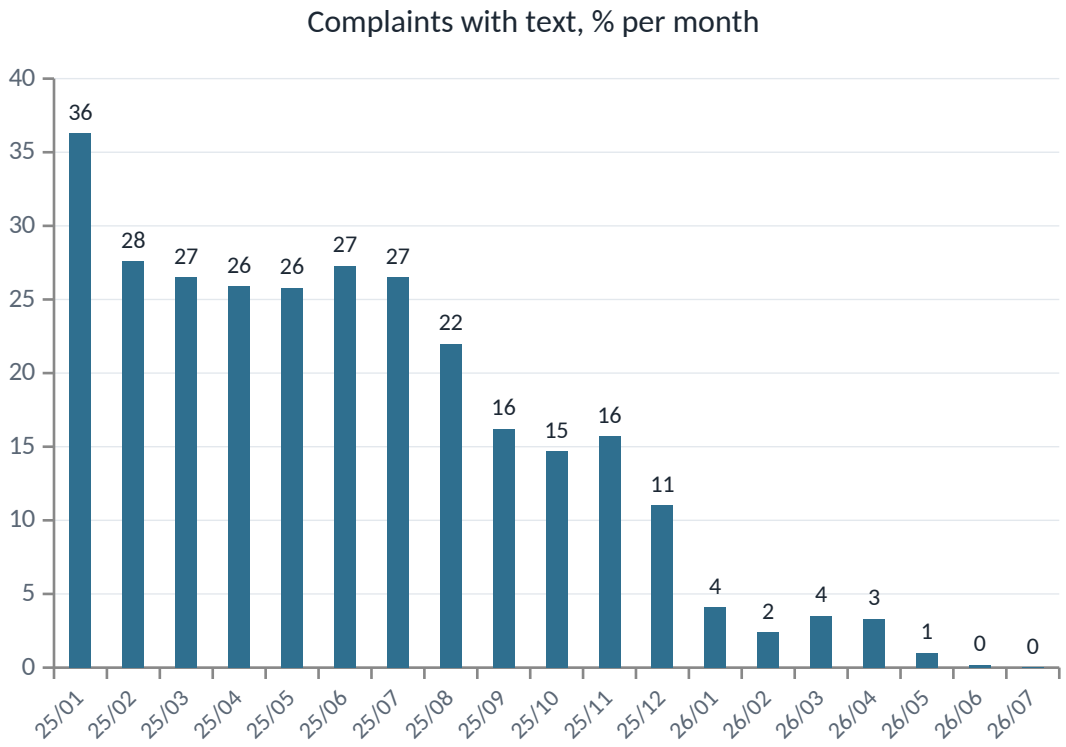
≥ 99.5%
availability

≤ 3 days
data freshness

Data sample: one real complaint per class

Class (label)	Company	Complaint text
Checking or savings	KeyCorp	*I am disputing a negative account reported by KeyBank that is blocking me from opening any bank accounts…*
Credit or prepaid card	Synchrony	*I submitted proof of product return as requested. I received letters from Synchrony Bank…*
Credit reporting	TransUnion	*The information listed below, which appears on my credit report, is the result of that…*
Debt collection	Experian	*There are collection accounts on my report that I believe contain inaccurate information…*
Debt or credit mgmt	TransUnion	*I requested proof of late payments. I requested the late payments be investigated…*
Money transfer	Block, Inc.	*I contacted Cash app and they refuse to give my money back even after I showed proof it was a scam…*
Mortgage	Shellpoint	*I have paid more than XXXX of my mortgage… but the Newrez refused to cancel my PMI payment…*
Payday / personal loan	Regional Mgmt	*They changed the approved loan amount & terms 3 different times…*
Student loan	MOHELA	*Errors in loan balances, misapplied payments, and wrongful denials of repayment plans…*
Vehicle loan or lease	Ford Credit	*I set this account up on autopay… for some reason it was not set…*

3.8 M usable texts; CFPB stopped publishing text in 2026



3.8 M

complaint texts, 2015–2025

26% → 0%

share with text, 2025-07 → 2026-07

Design response

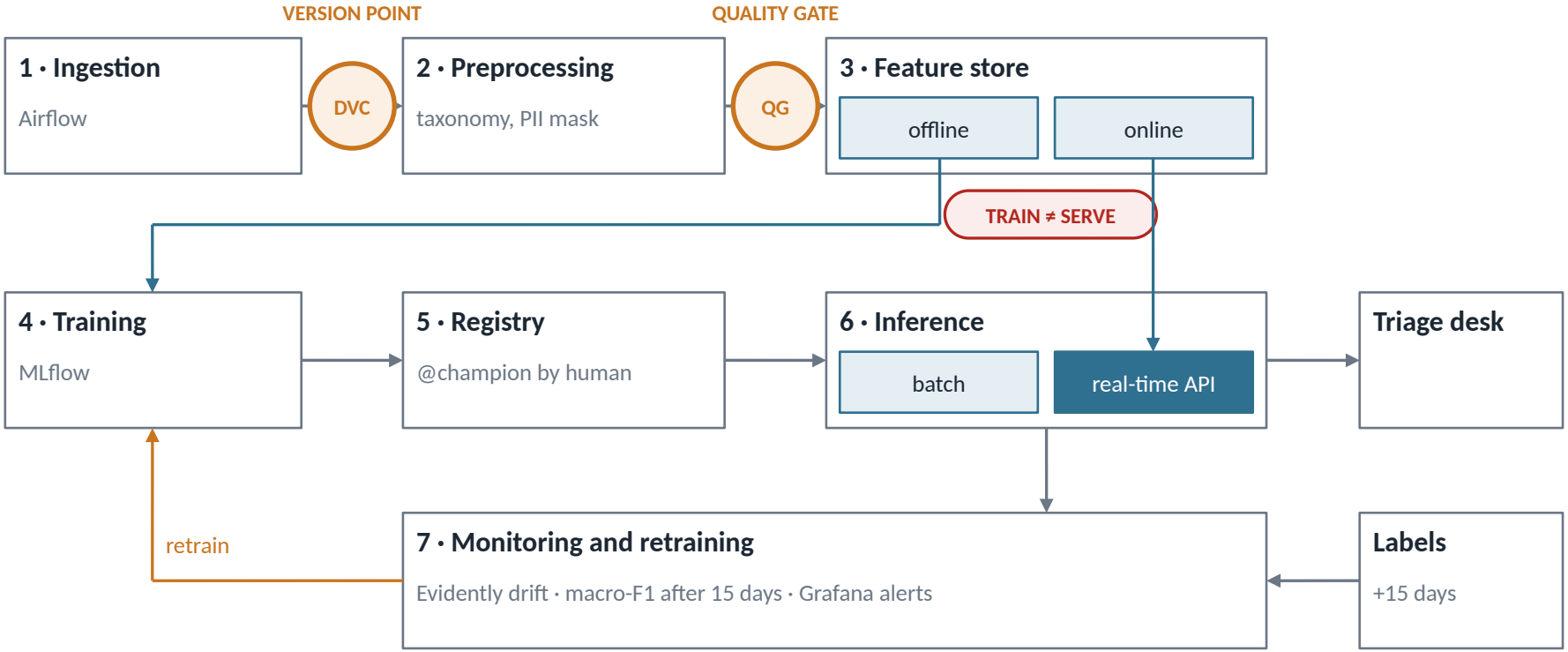
- Snapshot text in DVC
- Replay one month per run

The data passes all four checks the labs require

PASS	**Time axis**	daily dates since 2011
PASS	**History to hold out**	3.8 M texts; quarter-long val and test
PASS	**Label delay**	15 days (CFPB publication rule)
PASS	**Text corpus**	narratives + Ask CFPB + eCFR regulations



Data flow on the seven layers, with the three marks



This project carries all five labs

Lab	Needs	This project provides	Fit
1 · Design	design + data justification	design.md, data profile, sample, CI	Done
2 · Data + MLflow	batches, held-out period	monthly replay of 3.8 M texts	Fits
3 · LLM + retrieval	LLM baseline, corpus	zero-shot LLM vs TF-IDF; Ask CFPB + eCFR	Fits
4 · Serving + drift	servable model, real drift	FastAPI; 2023 taxonomy change, volume surge	Fits
5 · Agent	tasks to automate	drift review, model comparison, promotion note	Fits

Caveat: no new complaint text after 2025, so new batches are replayed history.